A note on our notation

Written by E. M. Wolkovich, with extensive help from T. J. Davies and V. Van der Meersch. Thanks also to Liam Revell for explaining basic BM notation.

In the paper Morales-Castilla *et al.* (2024) we present a hierarchical phylogenetic model using a common but confusing notation for the parameterization of the species × species matrix. Here I try to clarify and correct the notation a little.

The goal of the model is to understand how environmental cues (forcing, chilliing, photoperiod) affect budburst across species; I will just skip all this though and consider an intercept only model for simplicity.

Let's speed through the start of the model as the notation there is unchanged. For each observation i of species j, we assumed that the timing of phenological events were generated from the following sampling distribution:

$$y_{i,j} \sim \mathcal{N}(\mu_j, \sigma_e^2)$$

where

$$\mu_j = \alpha_j$$

and σ_e^2 represents random error unrelated to the phylogeny. And the species-specific intercept α_j is elements of the following normal random vectors:

$$\boldsymbol{\alpha} = [\alpha_1, \ldots, \alpha_n]^T \text{ such that } \boldsymbol{\alpha} \sim \mathcal{N}(\mu_\alpha, \boldsymbol{\Sigma_\alpha})$$

where the means of the multivariate normal distributions are root trait values (i.e., budburst date prior to evolving across a phylogenetic tree) and $\boldsymbol{\Sigma_\alpha}$ is an $n \times n$ phylogenetic variance-covariance matrix[†] of the form (**this is where it gets confusing**):

$$\begin{bmatrix} \sigma_i^2 & \lambda_i \times \sigma_i \times \rho_{12} & \ldots & \lambda_i \times \sigma_i \times \rho_{1n} \\ \lambda_i \times \sigma_i \times \rho_{21} & \sigma_i^2 & \ldots & \lambda_i \times \sigma_i \times \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda_i \times \sigma_i \times \rho_{n1} & \lambda_i \times \sigma_i \times \rho_{n2} & \ldots & \sigma_i^2 \end{bmatrix}$$

where σ_i^2 is the rate of evolution across a tree for a given trait or predictor (here assumed to be constant along all branches), λ_i scales branch lengths and therefore is a measure of the phylogenetic signal or extent of phylogenetic relatedness on each model parameter (i.e., α_j), and ρ_{xy} is the phylogenetic correlation between species x and y, or the fraction of the tree shared by the two species.

What's confusing here?

A few things. First, the i notation mixed with the numbers on ρ_{xy} is not really helpful, and it's made worse by our choice to use both σ^2 and σ notation here (and you may now notice—we never defined those σ). Here σ^2 is the Brownian motion rate of evolution across the tree, and the other σ should also be squared (see below 'Want even more information?' for more details). So the above works if you drop the i notation (it never should have been on σ^2 or λ in my estimation as these are single parameters estimated across the entire tree) and square all the σ as shown below:

$$\begin{bmatrix} \sigma^2 & \lambda \times \sigma^2 \times \rho_{12} & \ldots & \lambda \times \sigma^2 \times \rho_{1n} \\ \lambda \times \sigma^2 \times \rho_{21} & \sigma^2 & \ldots & \lambda \times \sigma^2 \times \rho_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \times \sigma^2 \times \rho_{n1} & \lambda \times \sigma^2 \times \rho_{n2} & \ldots & \sigma^2 \end{bmatrix}$$

We could have also maybe have written this as:

$$\begin{bmatrix} \sigma^2 & \lambda \times \sigma_{12} & \ldots & \lambda \times \sigma_{1n} \\ \lambda \times \sigma_{21} & \sigma^2 & \ldots & \lambda \times \sigma_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \times \sigma_{n1} & \lambda \times \sigma_{n2} & \ldots & \sigma^2 \end{bmatrix}$$

Above I have dropped ρ (it's just inherently in there, apparently this is a thing people do). Now σ^2 is still the Brownian motion rate of evolution across the tree, and $\sigma_{i,j}$ is the covariance between representative taxa (i and j). This is not an uncommon representation but: (1) we actually used the correlation matrix here and I find it bizarre to leave out, and (2) I think we could have been clearer if we presented the last part of our notation as:

[†]Phylogenetic people always call this the variance-covariance matrix, but if you want to look it up on Wikipedia, you can look for the covariance matrix, where you will see an additional four names for this sort of matrix. Yay, terminology!

$$\mathbf{\Sigma}_\alpha = \sigma^2_{BM} \cdot \begin{bmatrix} \rho_{1,1} & \lambda \cdot \rho_{1,2} & \dots & \lambda \cdot \rho_{1,n} \\ \lambda \cdot \rho_{2,1} & \rho_{2,2} & \dots & \lambda \cdot \rho_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \cdot \rho_{n,1} & \lambda \cdot \rho_{n,2} & \dots & \rho_{n,n} \end{bmatrix}$$

Here I moved σ^2 to outside the matrix since it a common factor multiplied across elements of the matrix (and I added $_{BM}$ to denote Brownian Motion). Then I show the ρ across all elements also, even the diagonals (and I use the dot multiplier throughout, and I add a comma in the subscript when referencing elements, both of which I find easier to read).

By definition $\rho_{1,1} = 1$ so the above also equals:

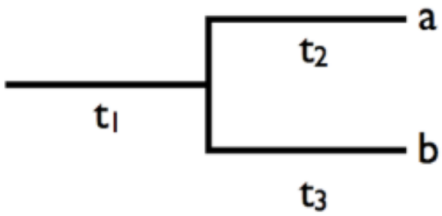
$$\mathbf{\Sigma}_\alpha = \sigma^2_{BM} \cdot \begin{bmatrix} 1 & \lambda \cdot \rho_{1,2} & \dots & \lambda \cdot \rho_{1,n} \\ \lambda \cdot \rho_{2,1} & 1 & \dots & \lambda \cdot \rho_{2,n} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \cdot \rho_{n,1} & \lambda \cdot \rho_{n,2} & \dots & 1 \end{bmatrix}$$

As you can now see, λ is a multiplier of the off-diagonal elements. Estimates of $\lambda = 1$ essentially leave the matrix untransformed and suggest that the species intercepts (or β etc.) are completely consistent with Brownian motion, while estimates of $\lambda = 0$ suggest no phylogenetic patterning according to Brownian motion. Further, when $\lambda = 0$ the model effectively defaults to a normal distribution, and thus simple partial pooling (or 'random effects' if you really want to call it that).

Want even more information?

You can also check out Pearse *et al.* (2025), where we tried to do the notation much better in explaining this for variance-covariance matrices. If you want even more, I have included below a basically verbatim version of information kindly sent from Liam Revell:

First, this is from Chapter 3 of Luke's book (Harmon, 2019).



Here the variance of taxon a might be denoted $\sigma(a)^2$ and under our Brownian process the value of $\sigma(a)^2$ would be $\sigma(B)^2 * (t1 + t2)$ in which $\sigma(B)^2$ is the Brownian rate. (It's given using the symbol for a variance because it's specifically the rate at which variance accumulates under our stochastic process.) The covariance between taxa a and b we might give as $\sigma(a, b)$ and in this case it'll have a value of $\sigma(B)^2 * t1$ – the rate of evolution multiplied by the evolutionary history shared between the two taxa, a and b.

If we jump to Chapter 6 of Luke's book then we see:

$$\mathbf{C_o} = \begin{bmatrix} \sigma^2_1 & \sigma_{12} & \dots & \sigma_{1r} \\ \sigma_{21} & \sigma^2_2 & \dots & \sigma_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \sigma_{r1} & \sigma_{r2} & \dots & \sigma^2_r \end{bmatrix}$$

This is just the covariance matrix... so σ^2_1 is the variance for taxon 1, which we'd calculate under Brownian motion as $\sigma(B)^2$ times the sum of the edge lengths leading from the root of the tree to the taxon 1 (or equivalently, the height above the root of taxon 1); σ_{12} is the covariance of taxa 1 and 2 which we'd calculate as $\sigma(B)^2$ times the height above the root of the MRCA (most recent common ancestor) of taxa 1 and 2; and so on.

When introducing lambda, we can think of it as a transformation of the covariance matrix in which all non- diagonal elements are multiplied by lambda.

$$\mathbf{C_\lambda} = \begin{bmatrix} \sigma^2_1 & \lambda \cdot \sigma_{12} & \dots & \lambda \cdot \sigma_{1r} \\ \lambda \cdot \sigma_{21} & \sigma^2_2 & \dots & \lambda \cdot \sigma_{2r} \\ \vdots & \vdots & \ddots & \vdots \\ \lambda \cdot \sigma_{r1} & \lambda \cdot \sigma_{r2} & \dots & \sigma^2_r \end{bmatrix}$$

However, since $\sigma(B)^2$ (the rate of Brownian evolution) is a common factor of all the elements in C, we

could equivalently first factor out $\sigma(B)^2$, and then multiply what's left (the times above the root of the MRCA of each $_{i,jth}$ pair of taxa) by λ. The result is the same. (This is why lambda is sometimes also represented as a tree transformation.)

References

Harmon, L.J. (2019) *Phylogenetic comparative methods*. Independent Traverse City, MI.

Morales-Castilla, I., Davies, T., Legault, G., Buonaiuto, D., Chamberlain, C.J., Ettinger, A.K., Garner, M., Jones, F.A., Loughnan, D., Pearse, W.D. *et al.* (2024) Phylogenetic estimates of species-level phenology improve ecological forecasting. *Nature Climate Change* pp. 1–7.

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